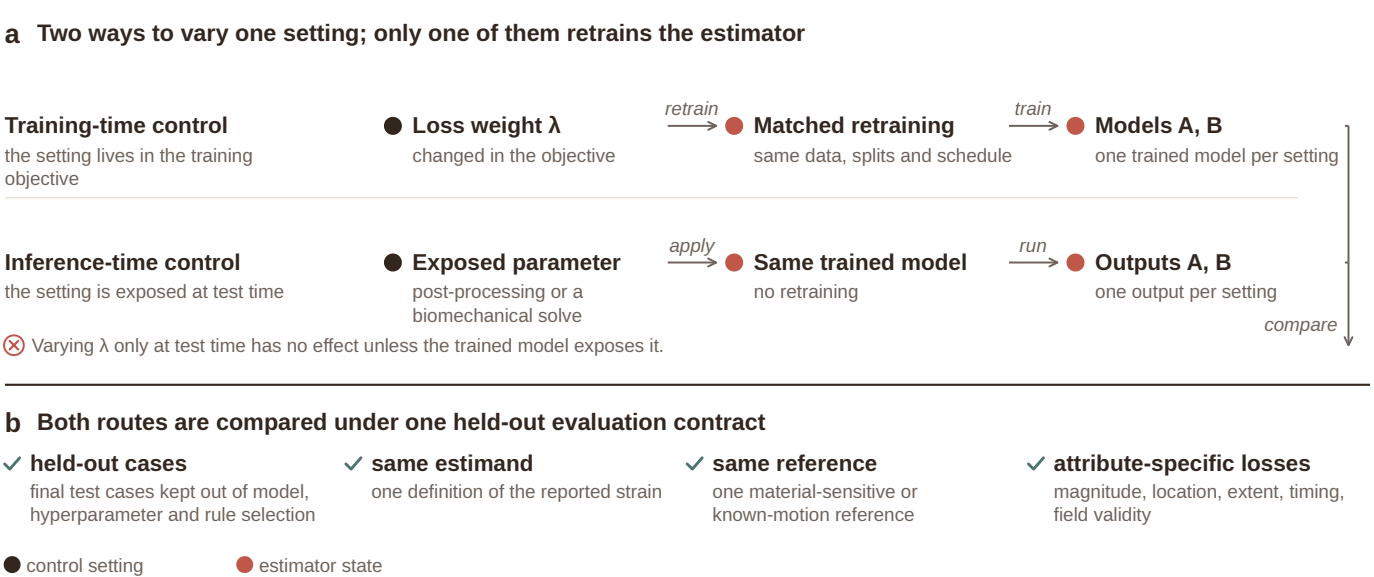
a Two ways to vary one setting; only one of them retrains the estimator
Training-time control
the setting lives in the training objective
Loss weight λ
changed in the objective
retrain
Matched retraining
same data, splits and schedule
train
Models A, B
one trained model per setting
Inference-time control
the setting is exposed at test time
Exposed parameter
post-processing or a biomechanical solve
apply
Same trained model
no retraining
run
Outputs A, B
one output per setting
compare
Varying λ only at test time has no effect unless the trained model exposes it.
b Both routes are compared under one held-out evaluation contract
held-out cases
final test cases kept out of model, hyperparameter and rule selection
same estimand
one definition of the reported strain
same reference
one material-sensitive or known-motion reference
attribute-specific losses
magnitude, location, extent, timing, field validity
control setting
estimator state